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ABSTRACT

Aim :

Our project aims to develop an efficient, local desktop application for library management.

Objective:

Replace outdated paper or Excel methods with user-friendly, digital systems.

Technologies:

Java (Swing for GUI), MySQL for Database Management, NetBeans IDE..

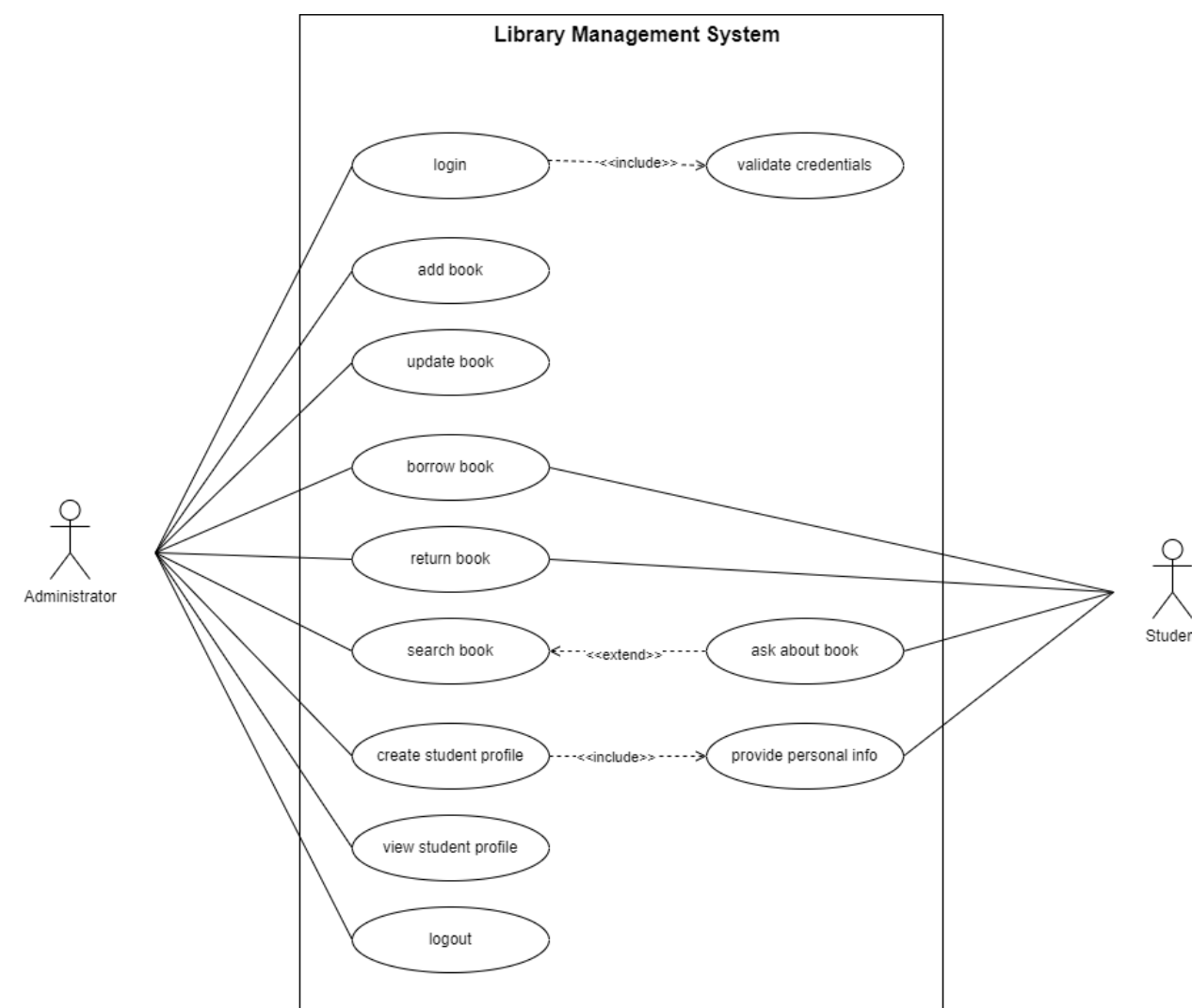
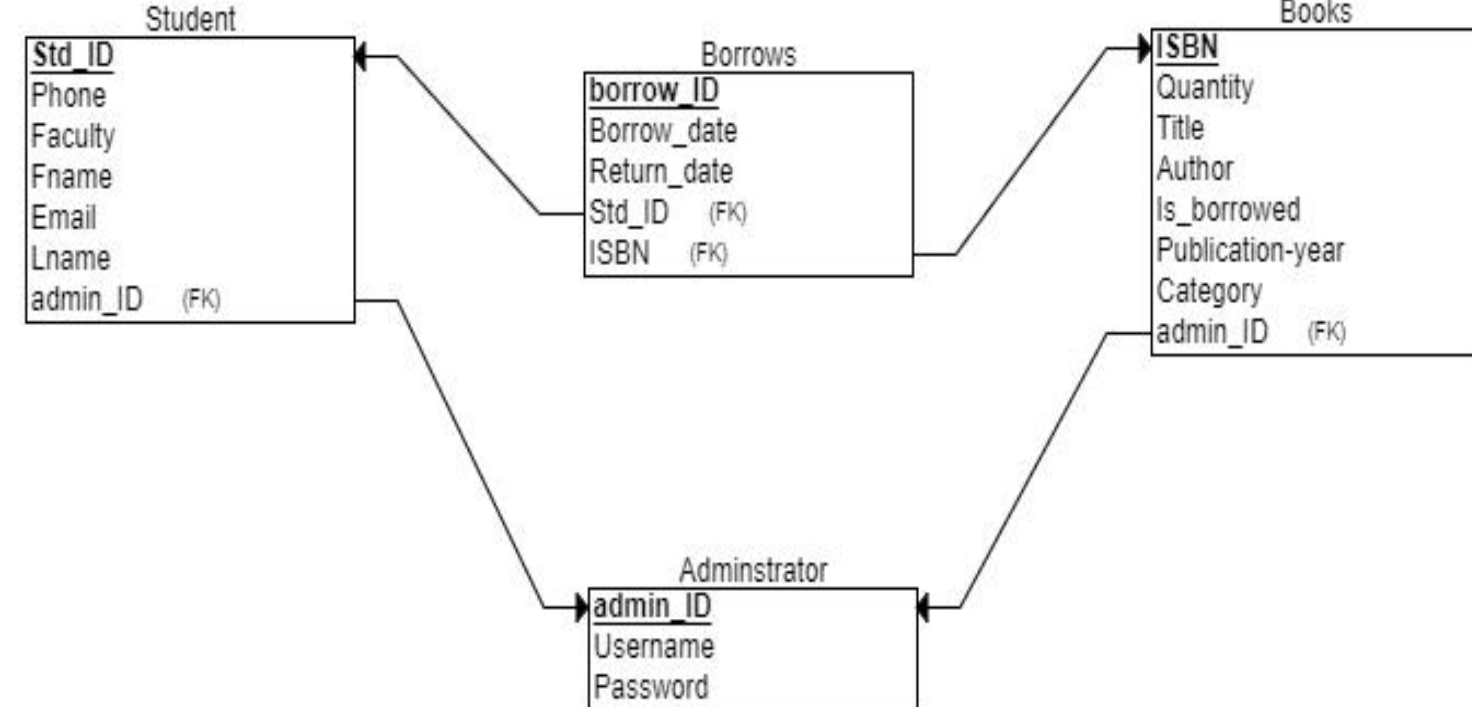
SYSTEM OVERVIEW

GUI Features:

- Secure admin login
- Book and Student Management
- Borrowing/Return system

Database Integration

- Reliable MySQL for data storage and retrieval
- Great performance, scalability and reliability
- Data integrity and consistency



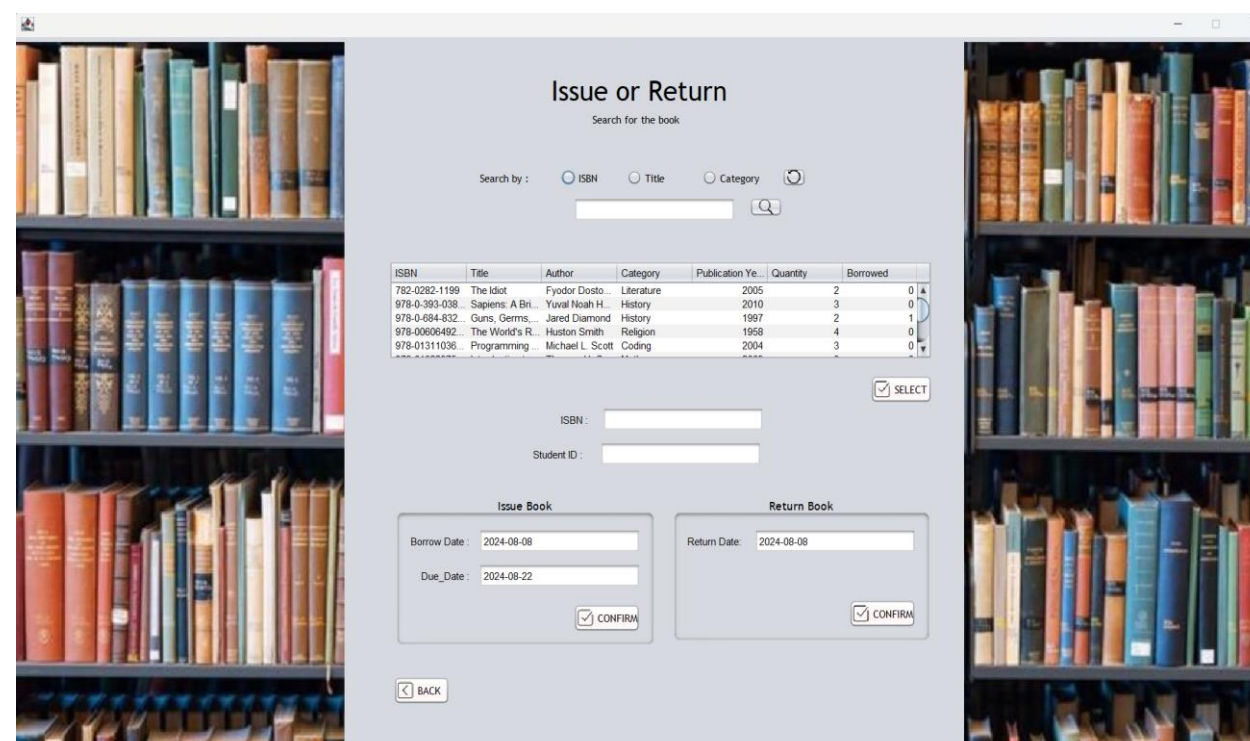
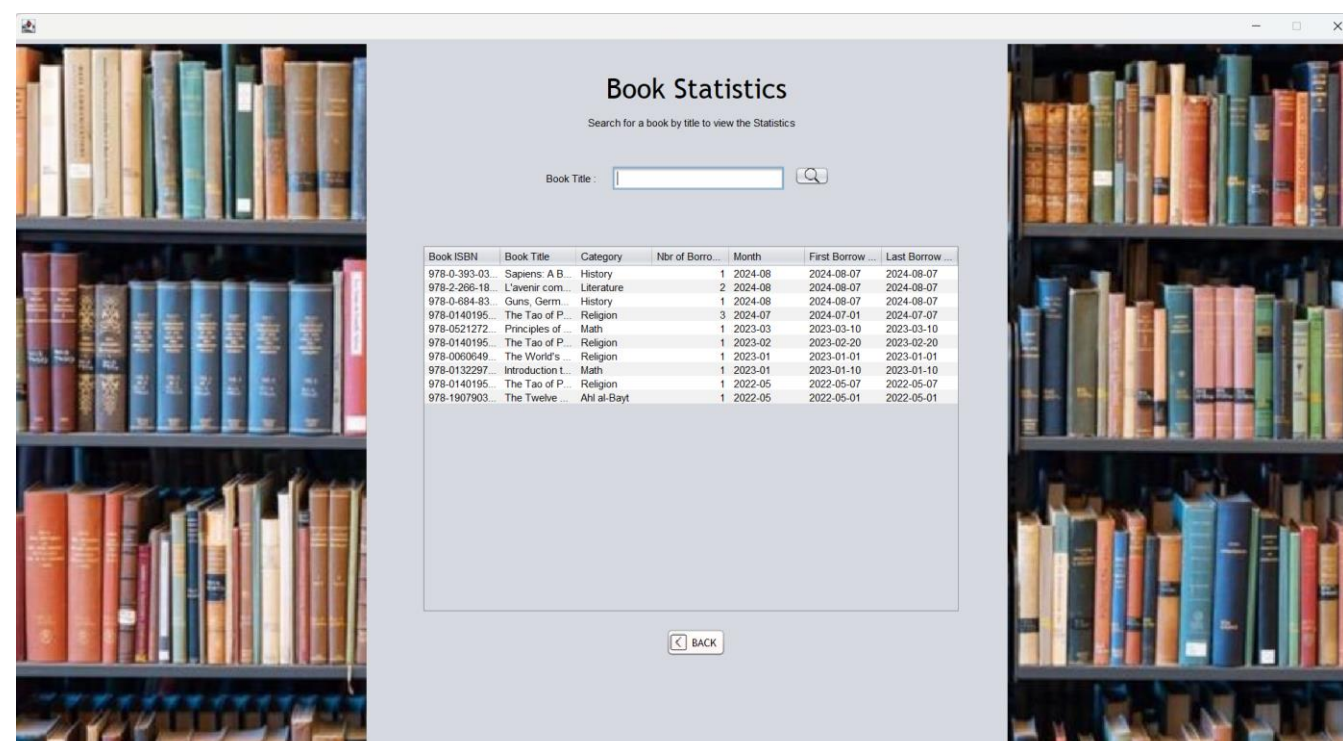
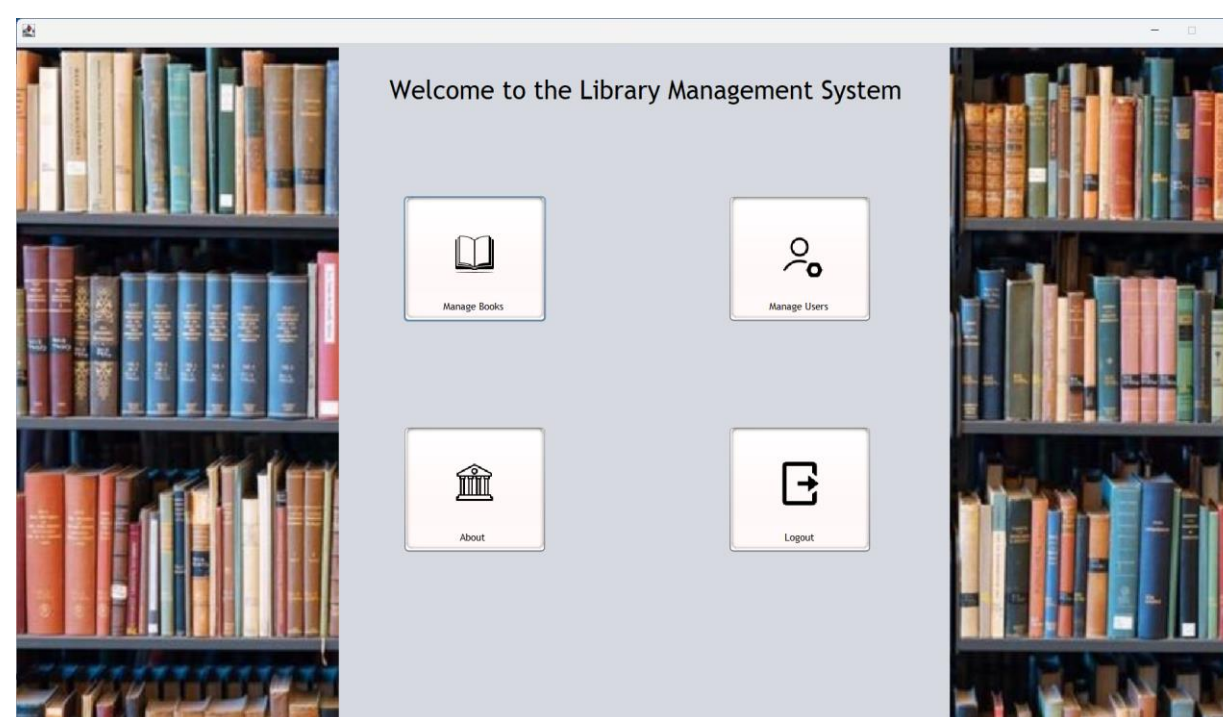
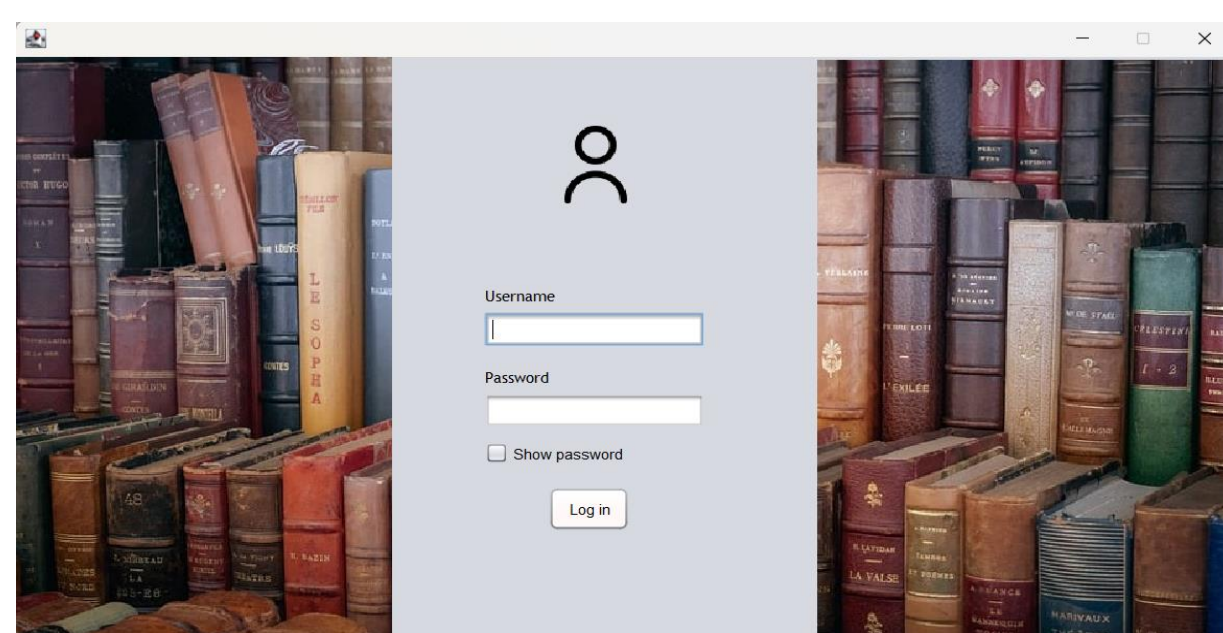
SOFTWARE IMPLEMENTATION

JAVA classes :

- About.java
- Book.java
- Student.java
- Dashboard.java
- Stats_Board.java
- ...

Database Connectivity

- JDBC
- SQL queries
- CRUD operations
- Statements and Prepared Statements



KEY FUNCTIONALITIES

• Login :

Secure admin login system.

• Book Management:

Add, edit, delete, and search books. Students can easily borrow books and the system automatically updates the book status, availability, and due date.

• Student Profiles

Administrators can create student profiles, and access detailed students' borrowing history.

• Borrowing and Returning Books

Track Student borrowings with a 2-week return period.

DATABASE STRUCTURE

Tables:

- Admin
- Student
- Books
- Borrowings

Views:

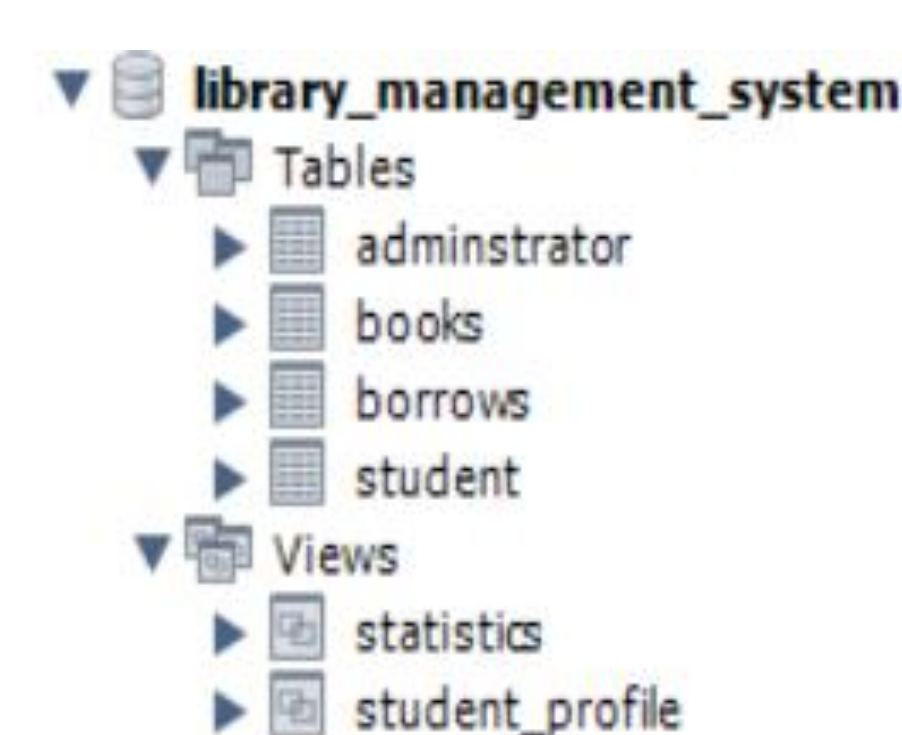
- Book Statistics
- Student Profiles

Triggers:

- Automated data updates.

	Field	Type	Null	Key	Default	Extra
▶	admin_ID	char(3)	NO	PRI	NULL	
	Username	varchar(50)	NO		NULL	
	Password	varchar(50)	NO		NULL	

	Field	Type	Null	Key	Default	Extra
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	Field	Type	Null	Key	Default	Extra
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	Faculty	varchar(50)	NO		NULL	
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	Phone	varchar(20)	NO		NULL	
	Faculty	varchar(50)	NO		NULL	
	admin_ID	char(3)	NO	MUL	NULL	

BENEFITS

- Offline capability: Works without a Wi-Fi connectivity
- Cross-platform compatibility
- Intuitive interface for library administrators, making it easy to use
- Tailored for the unique needs of university libraries

CONCLUSION

The University Library Management System provides a comprehensive and user-friendly solution for modern library management, offering **efficient book and student management**, streamlined **borrowing and returning processes**, and robust **reporting capabilities**. With its **intuitive interface**, **cross-platform compatibility**, and **secure data management**, the system is designed to meet the unique needs of university library administrators, empowering them to optimize library operations and enhance the overall user experience for students and faculty.